

Stability Analysis of Constant Power Load and Load Voltage Control Method for Wireless In-Wheel Motor

Daisuke Gunji¹, Takehiro Imura², and Hiroshi Fujimoto²

¹ The University of Tokyo / NSK Ltd., Japan

² The University of Tokyo, Japan

Abstract—In order to improve motion control performance of electric vehicles (EVs), In-Wheel Motor (IWM) is the most preferred electric motor arrangement. Our research group has proposed Wireless In-Wheel Motor (W-IWM) concept to solve some technical problems of an IWM, which are reliability and safety issues of power and signal wires. In this research, we did a stability analysis of Series-Series compensated wireless power transfer via magnetic resonance coupling with constant power load. Analysis result is verified by circuit simulation. Also, we propose a load voltage control method using a secondary converter. The effectiveness of the proposed control method has been verified by simulation and experiment using small power test equipment.

I. INTRODUCTION

Wireless power transfer (WPT) technology has received broad attention in recent years especially for automotive applications. Typical application of WPT for automotive are wireless power charging to electric vehicles (EVs) while driving [1], [2] and while parking [3], [4]. Magnetic resonance coupling [5] is the most preferred method of WPT for automotive applications because it can achieve high power transfer efficiency if transmitting distance is several tens of centimeters.

Advantages of EVs are not only environmental performance but also motion control [6]. In-Wheel Motor (IWM) is suitable structure because it can achieve high control bandwidth and independent drive of each wheel. However, there are reliability and safety issues because of power and signal wires. In order to solve that problem, our research group has proposed Wireless In-Wheel Motor (W-IWM) [7]. Concept of W-IWM is shown in Fig. 1. Electric power is bidirectionally transmitted from the primary side (on-board side) to the secondary side (in-wheel side). Signals are also communicated by wireless protocol. Then, completely wireless structure can be achieved. W-IWM is also applicable for dynamic wireless power supply from coils which are set under road.

In the case of W-IWM, electrical load is permanent magnet synchronous motor which is driven by a voltage type PWM inverter. Then, the load is considered as a constant power load. A number of previous researches exist about constant power load on power electronics field. In a DC-DC converter application, a constant power load shows unstable behavior because it has negative resistance characteristics [8]. Then, stability analysis is needed in the W-IWM application.

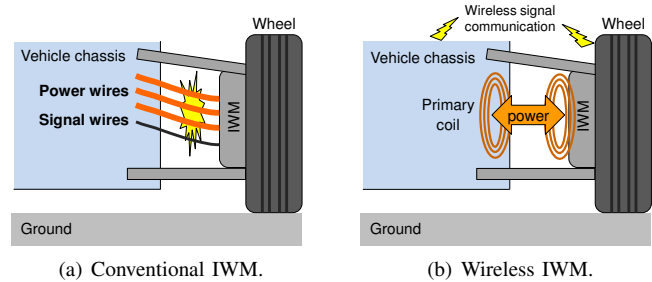


Fig. 1. Concept of Wireless In-Wheel Motor.

In this research, we analyze stability of a constant power load on wireless power transfer by magnetic resonance coupling which is Series-Series compensated circuit using simplified circuit model. Analysis result shows a load voltage of a constant power load is unstable. Analysis result is verified by circuit simulation and experiment. We also propose a load voltage stabilization control method. The effectiveness of the proposed method is verified by simulation and experiment using small power test equipment.

II. WIRELESS IN-WHEEL MOTOR (W-IWM)

A. Structure

Circuit structure of the W-IWM is shown in Fig. 2. Both the primary and the secondary resonance capacitors are series connection to coils (Series-Series compensated). The primary side (on-board side) consists of a buck-boost converter and a full-bridge inverter. Battery voltage is converted to high frequency which is same as the resonance frequency using the primary inverter. The electric power is transmitted to the secondary side (in-wheel side) via magnetic resonance coupling. Received electric power is converted to DC by a secondary converter which is also full-bridge circuit. Finally, a PWM inverter drives a permanent magnet synchronous motor (PMSM).

Some previous researches use a full-bridge diode rectifier and a DC-DC converter on the secondary side [9], [10]. However that circuit structure is not available for bi-directional power transfer and downsizing. On the other hand, the above circuit structure is capable of bi-directional power transfer

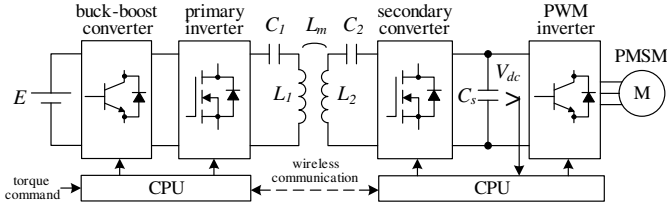


Fig. 2. Circuit structure of W-IWM.



(a) Experimental vehicle.



(b) First trial unit.

Fig. 3. Experimental vehicle and first trial unit.

because circuit structure is symmetric between the primary side and the secondary side.

B. First trial unit

Fig. 3(a) shows the experimental electric vehicle *FPEV4-Sawyer* which has developed by our research group. Drive unit of the vehicle can be easily exchanged. Then, it is possible to compare some types of drive units on the same vehicle.

We are now developing a first trial unit of W-IWM which is shown in Fig. 3(b). Final target of electrical power is 48 kW using four wheels, but in the first trial unit, maximum power is 6.6 kW using two wheels. By using the first trial unit, we establish control methods of power conversion circuit, and coil design.

III. STABILITY ANALYSIS OF THE LOAD VOLTAGE

A. Simplified circuit model

A SS-compensated WPT circuit can be expressed using T-type equivalent circuit as shown in Fig. 4 [11]. Output voltage of the primary inverter is actually square wave. In this research, we focus on the fundamental wave components v_{11} . In the similar way, we focus on the fundamental wave components of the secondary converter input voltage v_{21} . In addition, we assume that the secondary current i_{21} is sinusoidal wave and in phase with v_{21} .

The impedance matrix of the boxed circuit on Fig. 4 is expressed as follows:

$$\mathbf{Z} = \begin{bmatrix} R_1 + \frac{1}{j\omega_0 C_1} + j\omega_0 L_{1m} & j\omega_0 L_m \\ j\omega_0 L_m & R_2 + \frac{1}{j\omega_0 C_2} + j\omega_0 L_{2m} \end{bmatrix}, \quad (1)$$

where ω_0 is angular frequency, $R_{1,2}$ are respectively resistance of coils, $C_{1,2}$ are respectively resonance capacitance, $L_{1,2}$ are respectively coil inductance, L_m is mutual inductance, $L_{1m} =$

TABLE I
SPECIFICATIONS OF W-IWM.

	Final target	first trial unit
Number of IWM	4	2
Max total power	48 kW	6.6 kW
Max total torque	1300 Nm	475 Nm

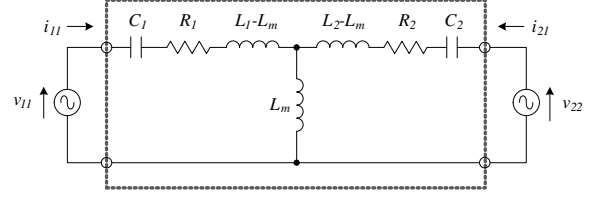


Fig. 4. T-type equivalent circuit of SS-compensated WPT circuit.

$L_1 - L_m$, and $L_{2m} = L_2 - L_m$. We assume that the circuit meets following resonance condition:

$$\omega_0 = \frac{1}{\sqrt{L_1 C_1}} = \frac{1}{\sqrt{L_2 C_2}}. \quad (2)$$

Then, $\mathbf{Z}$ is expressed as follows:

$$\mathbf{Z} = \begin{bmatrix} R_1 & j\omega_0 L_m \\ j\omega_0 L_m & R_2 \end{bmatrix}. \quad (3)$$

From eq.(3), the secondary current i_{21} is calculated as follows:

$$\begin{bmatrix} i_{11} \\ i_{21} \end{bmatrix} = \mathbf{Z}^{-1} \begin{bmatrix} v_{11} \\ v_{21} \end{bmatrix}, \quad (4)$$

$$i_{21} = -\frac{j\omega_0 L_m v_{11} - v_{21}}{R_1 R_2 + (\omega_0 L_m)^2}. \quad (5)$$

Assuming that v_{21} is $\frac{\pi}{2}$ phase advance to v_{11} , RMS value of i_{21} is expressed as follows:

$$I_{21} = \frac{\omega_0 L_m V_{11} - R_1 V_{21}}{R_1 R_2 + (\omega_0 L_m)^2}, \quad (6)$$

where V_{11} is RMS value of v_{11} , and V_{21} is RMS value of v_{21} . If the secondary converter is operated as full-bridge rectifier, average output current of the secondary converter i_s is expressed as follows:

$$i_s = \frac{2\sqrt{2}}{\pi} \frac{\omega_0 L_m V_{11} - R_1 V_{21}}{R_1 R_2 + (\omega_0 L_m)^2}. \quad (7)$$

Generally, $\omega_0 L_m$ is much bigger than R_1 , it can be assumed that i_s is constant current source at a operating point.

If the secondary converter is operated as full-bridge rectifier, input voltage of the secondary converter is square wave. Focus on the fundamental wave component of that, RMS value of the secondary converter input voltage V_{21} is expressed by the following equation.

$$V_{21} = \frac{2\sqrt{2}}{\pi} v_L, \quad (8)$$

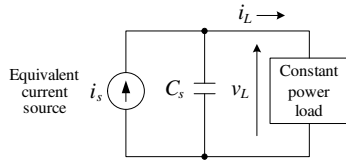


Fig. 5. Simplified circuit of the secondary side.

where v_L is the load voltage. At a operating point, equivalent load resistance of the constant power load is expressed as follows:

$$R_L = \frac{v_L}{i_L} = \frac{v_L^2}{p_L}, \quad (9)$$

where i_L is the load current, and p_L is the load power. The simplified circuit of the secondary side is shown in Fig. 5.

B. Stability analysis

In this section, we derive the transfer function of the simplified circuit, and then analyze the stability of the circuit.

Circuit equation of the simplified circuit is expressed as follows:

$$i_L = I_s - C_s \frac{dv_L}{dt}, \quad (10)$$

where C_s is capacitance of the smoothing capacitor. The load current i_L is expressed as follows because the load is constant power load.

$$i_L = \frac{p_L}{v_L}. \quad (11)$$

By substituting eq.(11) into eq.(10), the following equation is derived.

$$\frac{dv_L}{dt} = -\frac{p_L}{C_s v_L} + \frac{i_s}{C_s}. \quad (12)$$

By linearizing eq.(12) around equilibrium point, the following equation is derived.

$$\begin{aligned} \frac{d\Delta v_L}{dt} &= \frac{p_L \Delta v_L}{C_s V_L^2} + \frac{\Delta i_s}{C_s}, \\ v_L &= V_L + \Delta v_L, \\ i_s &= I_s + \Delta i_s. \end{aligned} \quad (13)$$

Then, the transfer function $P_\Delta(s)$ from Δi_s to Δv_L is derived from Laplace transform of eq.(13):

$$P_\Delta(s) = \frac{\Delta v_L(s)}{\Delta i_s(s)} = \frac{1}{C_s \left(s - \frac{p_L}{C_s V_L^2} \right)}. \quad (14)$$

The pole of the $P_\Delta(s)$ is positive value in any case because $p_L > 0$ and $V_L > 0$. Then, $P_\Delta(s)$ is unstable system. From the above stability analysis, in the case of the constant power load, the load voltage of WPT is unstable. According to eq.(14), the pole becomes faster in any of the following cases:

- 1) Capacitance of the smoothing capacitor C_s is smaller.
- 2) The load power p_L is bigger.

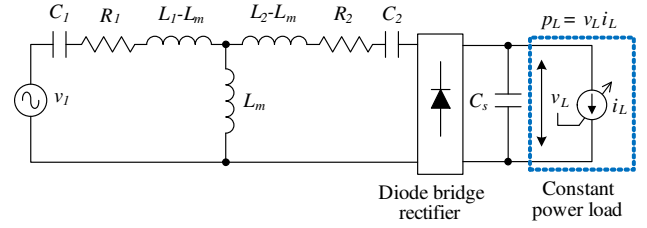


Fig. 6. Circuit simulation model.

TABLE II
CIRCUIT PARAMETERS.

	Primary	Secondary
Coil resistance $R_{1,2}$	0.547 Ω	0.535 Ω
Coil inductance $L_{1,2}$	166 μH	167 μH
Resonance capacitor $C_{1,2}$	19.9 nF	19.9 nF
Coil size	200 x 200 mm	
Coil gap	100 mm	
Mutual inductance L_m	21.8 μH	
coupling coefficient k	0.132	
Resonance frequency	87.6 kHz	
Smoothing capacitor C_s	1000 μF	

- 3) The operating load voltage V_L is smaller.

Therefore, the value of C_s is designed with not only the load voltage ripple but also the control performance and stability of the load voltage.

C. Circuit simulation

Circuit simulation has been performed to verify the stability analysis result using MATLAB Simulink SimPowerSystems. The simulation circuit model is shown in Fig. 6. The primary voltage is modeled by AC voltage source, the secondary converter is modeled as full-bridge diode rectifier, and the constant power load is modeled using controlled current source. Circuit parameters are listed in TABLE II. These parameters are same as the test equipment. Operating conditions and initial conditions are listed in TABLE III. The amplitude of the primary voltage is set by trial and error. Simulation is stopped when v_L close to zero because it is impossible to simulate constant power load.

Simulation results are shown in Fig. 7(a) and (b). Fig. 7(a) shows the load voltage v_L . Fig. 7(b) shows the load current i_L . According to Fig. 7(a) and (b), the load voltage and the load current are diverged. Compare blue lines on figures to red lines, time variation of the load voltage and current are completely different even though difference of v_1 is very small. In addition, v_2 decrease faster so that v_2 becomes small. That result agrees with stability analysis result. From above simulation results, it is shown that SS-compensated WPT circuit with a constant power load is unstable.

IV. LOAD VOLTAGE CONTROL METHOD

A. Operation of the secondary converter

In this chapter, we propose a load voltage feedback control method. The feedback loop is closed on the secondary side to prevent the influence of the signal communication delay between the primary side and the secondary side.

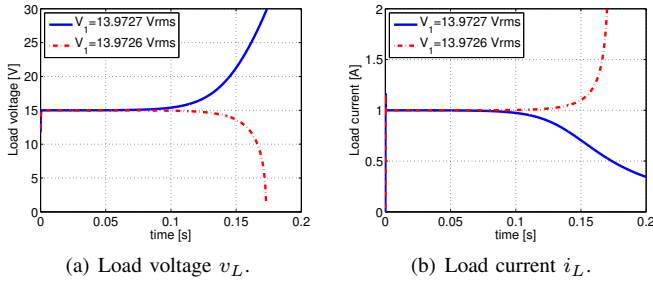


Fig. 7. Circuit simulation results.

TABLE III
SIMULATION CONDOTIONS.

Condition	Value
Load power	15 W
Initial load voltage	15 V
Initial load current	1.0 A
Operating frequency	87.6 kHz

In order to control the transmitting power on the secondary side, the secondary converter is operated with the following two operation modes:

- **Rectification mode:** All switching devices are OFF. The secondary converter is operated as full-bridge rectifier by diodes which are in parallel with switching devices as shown in Fig. 8 (a). Then, electric power is transmitted to the load. If the received power is bigger than the load power, the load voltage increases.
- **Short mode:** Both lower arm switching devices are ON. Electric power is not transmitted to the load because the secondary coil circuit is shorted as shown in Fig. 8 (b). Then, the load voltage decrease because the load power is supplied from the smoothing capacitor.

By switching above two operation modes depending on the load voltage, the load voltage can be controlled to a reference value. Hysteresis comparator is used to switch operation modes in previous research [12]. In this research, we propose another control method.

B. Load voltage control method

Assuming that the secondary current is sinusoidal wave and focus on the fundamental wave component, RMS value of the secondary current I_2 is approximated by the following equation:

$$I_2 \simeq \frac{\omega_0 L_m V_{11} - \frac{2\sqrt{2}}{\pi} R_1 V_L}{R_1 R_2 + (\omega_0 L_m)^2}, \quad (15)$$

where V_L is the load voltage. On each operation mode, output current of the secondary converter I_s are expressed as follows:

$$I_s = \begin{cases} \frac{2\sqrt{2}}{\pi} I_2 & \text{(Rectification mode)} \\ 0 & \text{(Short mode)} \end{cases} \quad (16)$$

One cycle time of above two operation mode is set to a constant time period. Assuming that the load voltage hardly

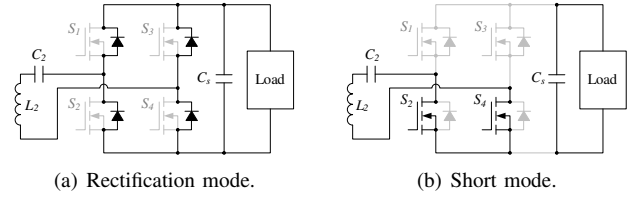


Fig. 8. Operation modes of the secondary converter.

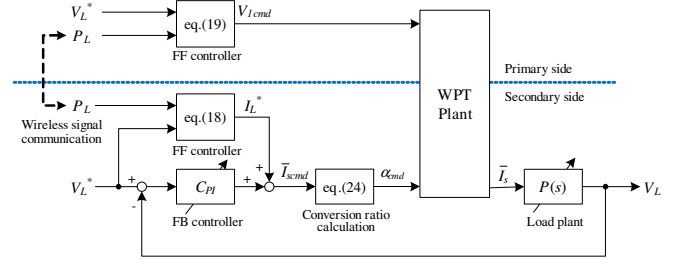


Fig. 9. Block diagram of the proposed control method.

changes between one period, average output current of the secondary converter $\bar{I}_s$ between one time period is expressed as follows:

$$\bar{I}_s = \frac{2\sqrt{2}}{\pi} I_2 \alpha, \quad (17)$$

where α is conversion ratio, that is time ratio of the rectification mode between one period. By manipulating α , the load voltage control can be achieved.

Block diagram of the proposed control method is shown in Fig. 9. The primary-side controller is a feedforward controller, and the secondary-side controller is a two-degrees-of freedom controller.

Relation between the load power and the load current reference I_L^* is expressed as follows:

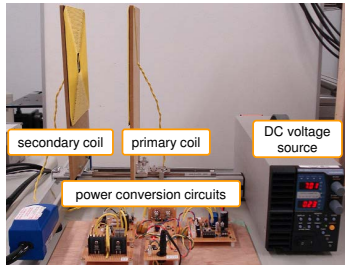
$$I_L^* = \frac{P_L}{V_L^*}, \quad (18)$$

where V_L^* is reference value of the load voltage. From eq.(17) and eq.(18), the primary voltage command V_{1cmd} is calculated as following equation:

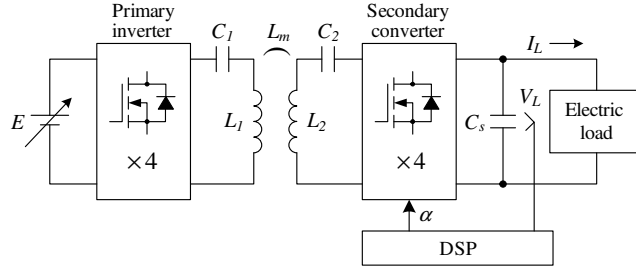
$$V_{1cmd} = \frac{\pi}{2\sqrt{2}} \frac{P_L \{R_1 R_2 + (\omega_0 L_m)^2\}}{\omega_0 L_m V_L^* \alpha_r} + \frac{2\sqrt{2}}{\pi} \frac{R_1 V_L^*}{\omega_0 L_m}, \quad (19)$$

where α_r is reference value of the α . That means, if there are no model error, transmitting power becomes same as the load power when $\alpha = \alpha_r$ and $V_{11} = V_{1cmd}$. Therefor, α indicates margin of the power transfer for model error, e.g. variation of the coupling coefficient due to coil position misalignment and gap variation.

The feedforward controller of the secondary side is same as eq.(18). The feedback controller of the secondary side is PI controller. The transfer function of the equivalent load plant



(a) Overview.



(b) Circuit structure.

Fig. 10. Experimental equipment.

is expressed as follows:

$$P(s) = \frac{R_L}{R_L C_s s + 1}, \quad (20)$$

where R_L is equivalent load resistance which is expressed as following equation.

$$R_L = \frac{V_L^{*2}}{P_L}. \quad (21)$$

Then, the PI controller is designed by pole placement method.

$$K_p = \frac{2pR_L C_s - 1}{R_L}, \quad (22)$$

$$K_i = p^2 C_s, \quad (23)$$

where K_p is proportional gain, K_i is integration gain, and p is closed loop poles (multiple root). Therefore, the feedback controller is variable with the load power.

Manipulated variable of the controller is average output current of the secondary converter $\bar{I}_{scmd}$. From eq.(15) and eq.(17), the conversion ratio command of the secondary converter α_{cmd} is calculated by following equation:

$$\alpha_{cmd} = \frac{\pi}{2\sqrt{2}} \frac{R_1 R_2 + (\omega_0 L_m)^2}{\omega_0 L_m V_{1cmd} - \frac{2\sqrt{2}}{\pi} R_1 V_L^*} \bar{I}_{scmd}. \quad (24)$$

V. SIMULATION AND EXPERIMENT

In this chapter, we verify the effectiveness of the proposed control method by simulation and experiment.

A. Experimental equipment

Experimental equipment is shown in Fig. 10(a). Circuit of the equipment is shown in Fig. 10(b). The equipment consists of a DC power supply, a primary inverter, primary and secondary coils, a secondary converter, an electric load (PLZ1004W: KIKUSUI), and a DPS (DS1104: dSPACE). Circuit parameters are listed in TABLE II. The load voltage is measured on the DSP, and conversion ratio of the secondary converter α is controlled by the proposed method. Primary voltage is set by changing DC supply voltage E .

TABLE IV
SIMULATION CONDITIONS.

Parameter	Value
Operation mode time period	2 ms
Load power P_L	2 W
Load voltage reference V_L^*	10 V
Conversion ratio reference α_r	0.7
Closed loop pole placement	8 Hz

B. Simulation

We did a simulation using MATLAB Simulink SimPowerSystems. Circuit model is same as Fig. 6 except for the secondary converter. On resistance of switching devices and forward voltage of body diodes are ignored. A first order low-pass filter is applied to the feedback value of V_L in order to suppress ripple. Cutoff frequency of the filter is 100 Hz. Other simulation conditions are listed in TABLE IV. In the case of without control, conversion ratio α is set to the average value of α which is with proposed control.

Simulation results are shown in Fig. 11(a) to (d). Fig. 11(a) shows time variation of the load voltage V_L . Without the proposed method, V_L decreased rapidly. On the other hand, the load voltage is successfully controlled to the reference value by the proposed method. Fig. 11(b) shows conversion ratio of the secondary converter α . Manipulation of α by the feedback controller is small, but that is significant to control V_L . Average value of α is bigger than α_r . It is considered that a transient response of the secondary current is the cause of the error. Fig. 11(c) shows close-up of V_L , and Fig. 11(d) shows the secondary converter output current I_s . According to these figures, electric power is received and V_L increases on the rectification mode, on the other hand, I_s is almost zero and V_L decreases on the short mode. The average value of the secondary converter output current is controlled to I_L^* by the proposed control method. From these simulation results, the effectiveness of the proposed method is verified.

C. Experiment

Experiment has been carried out using the experimental equipment. Experimental condition is same as the simulation condition as listed in TABLE IV. Cutoff frequency of the low-pass filter to suppress the measurement value of V_L is set to 20 Hz. On the experiment, feedforward value of the

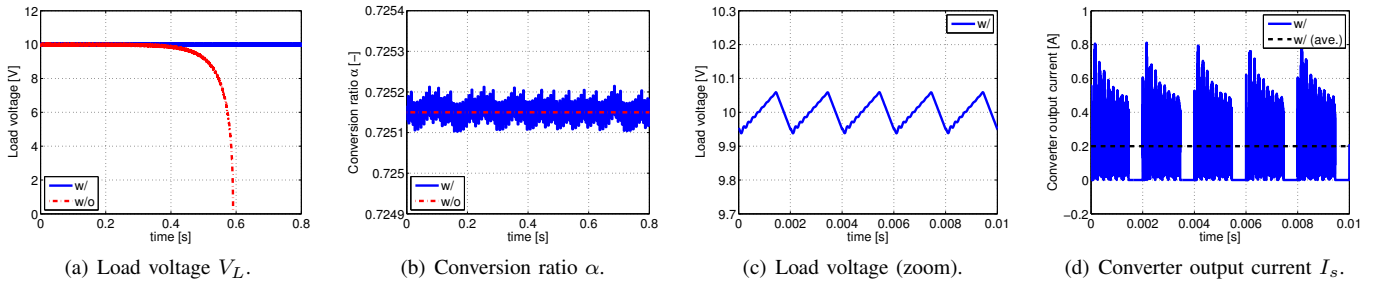


Fig. 11. Load voltage control simulation results.

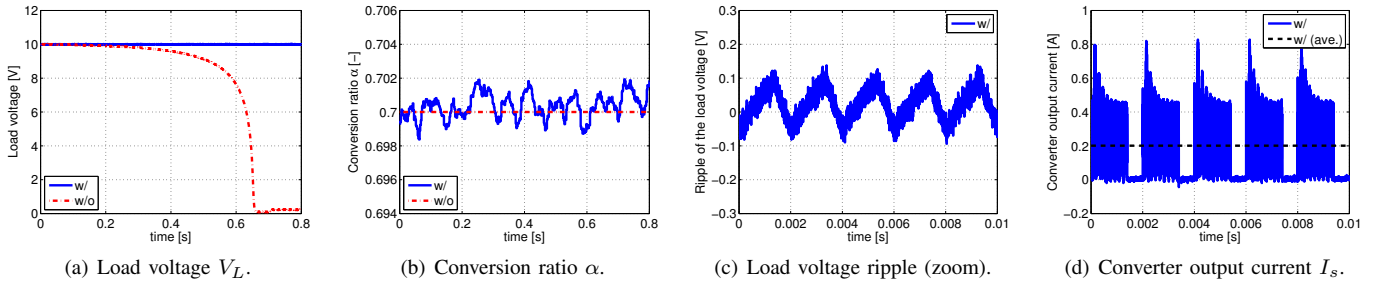


Fig. 12. Load voltage control experimental results.

primary voltage by eq.(19) is insufficient due to dead-time of the primary inverter, forward voltage of body diodes on the secondary converter, and modeling error. Therefore, we adjust the DC voltage E to be $\alpha = \alpha_r$ when with the proposed control.

Experimental results are shown in Fig. 12(a) to (d). These results are in good agreement with simulation results. The load voltage is successfully controlled to the reference value by the proposed method. According to Fig. 12(b), manipulation of α is little bigger than the simulation result. Fig. 12(c) shows ripple of the load voltage. Ripple width and time variation fit in well with the simulation result. Fig. 12(d) shows output current of the secondary converter. The current is successfully controlled by two operation modes.

From the above simulation and experimental results, the effectiveness of the proposed load voltage control method is verified.

VI. CONCLUSION

In this research, we did a stability analysis of SS-compensated WPT circuit with constant power load. According to the analysis result, that system is unstable. In order to stabilize the load voltage of the constant power load, we propose a load voltage feedback control method using the secondary converter. The effectiveness of the proposed method is verified by simulation and experiment. Future work is improvement of rectification method of the secondary converter.

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